AI SMART CAREER ROADMAP GENERATOR

Mini Project Report

Submitted by

Mokanpally Chandana

Pallavi Engineering College, Hyderabad

Department of Computer Science Engineering (Data Science)

Academic Year: 2026–2027

Learning Block 1 — Generative AI Application

Guided by: Raghavendra Rao M S

DECLARATION

I hereby declare that the mini project entitled “AI Smart Career Roadmap Generator” is prepared as an academic project. The proposed application and report are intended to demonstrate how Generative AI can support personalized career planning. Any implementation details, screenshots, and results should be verified against the working application before final submission.

Date: _____ Place: Hyderabad

Signature of Student: Mokanpally Chandhana

Mokanpally Chandhana

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1. Introduction

AI Smart Career Roadmap Generator is a proposed Generative AI–powered web application designed to help students explore career options and create a personalized learning plan. It uses details such as education, interests, existing skills, target role, and available study time to produce staged guidance.

The application aims to organize career preparation into manageable steps, including skills to learn, practice projects, learning resources, and milestones. AI-generated recommendations should be treated as guidance and reviewed against reliable course and job information.

- Personalized career suggestions based on a user profile.
- Skill-gap analysis for a selected target role.
- Step-by-step learning roadmap with milestones.
- Project ideas and learning-resource suggestions.
- Progress checklist to track completed activities.

2. Problem Statement

Students may find it difficult to choose a career direction and understand which skills to learn first. Advice is spread across websites, course platforms, and job descriptions. Without a structured plan, learners may struggle to connect their current skills with the requirements of a target role.

The proposed system brings profile information and AI-assisted suggestions into one workflow. It aims to generate a clearer learning sequence while reminding users to verify important recommendations.

3. Objectives

- Develop a simple web application for personalized career-roadmap generation.
- Collect education, interests, skills, target role, and weekly study time.
- Use a Generative AI model/API to suggest learning stages and relevant activities.
- Identify possible skill gaps and recommend practice projects.
- Display milestones in a clear format and allow progress tracking.
- Test the clarity and relevance of recommendations with sample profiles.

4. Project Scope

Target users: students, fresh graduates, and beginners exploring a career direction.

Included features

- Career-goal and profile input form.
- Suggested skills and skill-gap overview.
- Learning roadmap divided into stages.
- Practice project and resource recommendations.
- Milestone checklist and progress tracking.

Current boundaries

The quality of recommendations depends on the information supplied and the configured AI service. Job-market needs and course details can change. The initial version is a guidance tool and does not guarantee employment or replace advice from qualified career counselors.

5. Proposed System / Methodology

- User enters education, interests, current skills, target job, and available study time.
- The backend validates the profile and constructs a structured prompt.
- The prompt is sent securely to the configured Generative AI API.
- The returned suggestions are checked and formatted into stages, skills, projects, and milestones.
- The roadmap is displayed on a dashboard where the user can review recommendations and track progress.

6. System Architecture / Workflow

Stage	Process
1. User Interface	User enters profile and target career.
2. Input Processing	Validate and organize the submitted fields.
3. Prompt Builder	Create structured instructions using the profile.
4. GenAI Layer	Generate possible skills, projects, and learning stages.
5. Output Layer	Display roadmap, resources, and progress checklist.

Workflow: User Profile → Validation → Prompt Construction → GenAI API → Response Formatting → Roadmap Dashboard

7. Implementation

Suggested technologies and tools

Technology	Purpose
HTML, CSS, JavaScript	Responsive user interface and forms
Python with Flask/FastAPI	Backend logic and API routing
GenAI API	Generate personalized roadmap suggestions
SQLite (optional)	Store profiles, saved roadmaps, and progress
VS Code / Browser	Development and testing

Implementation note: This report describes a proposed design. Update the stack to match the technologies actually used in the working application. Keep API keys on the backend and avoid collecting unnecessary personal information.

8. User Interface / Application Screenshots

Figure 1: Home / Main Dashboard

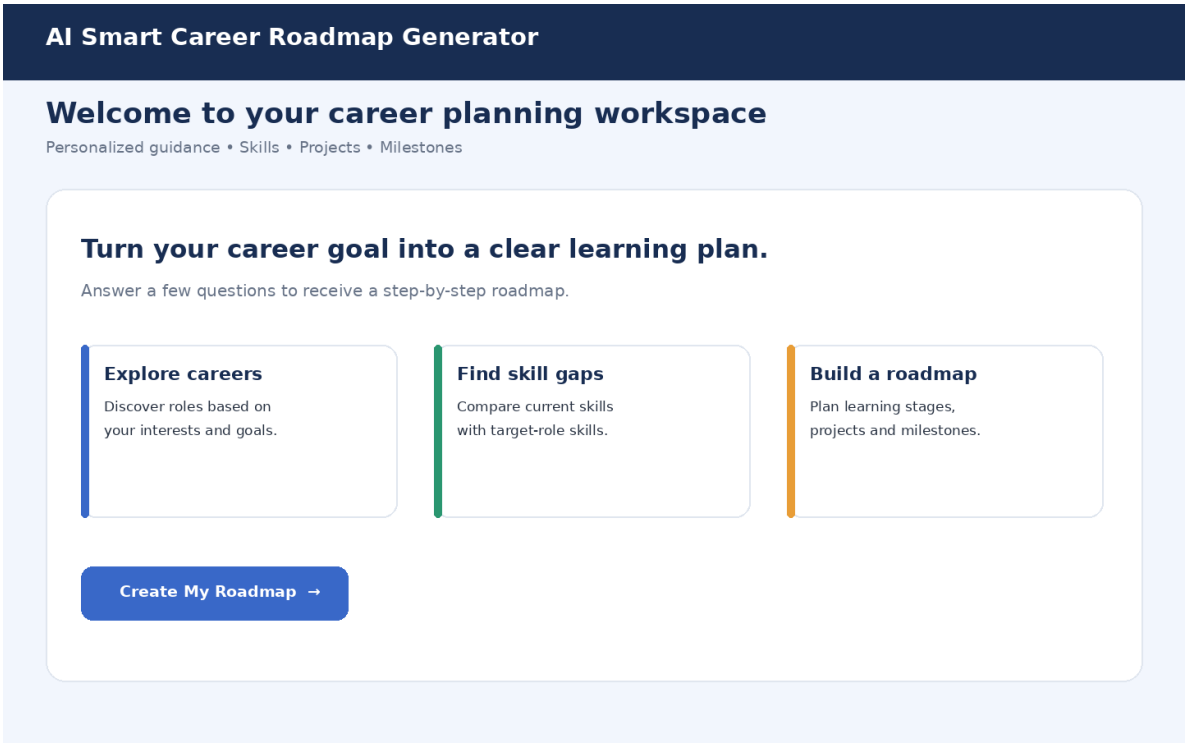
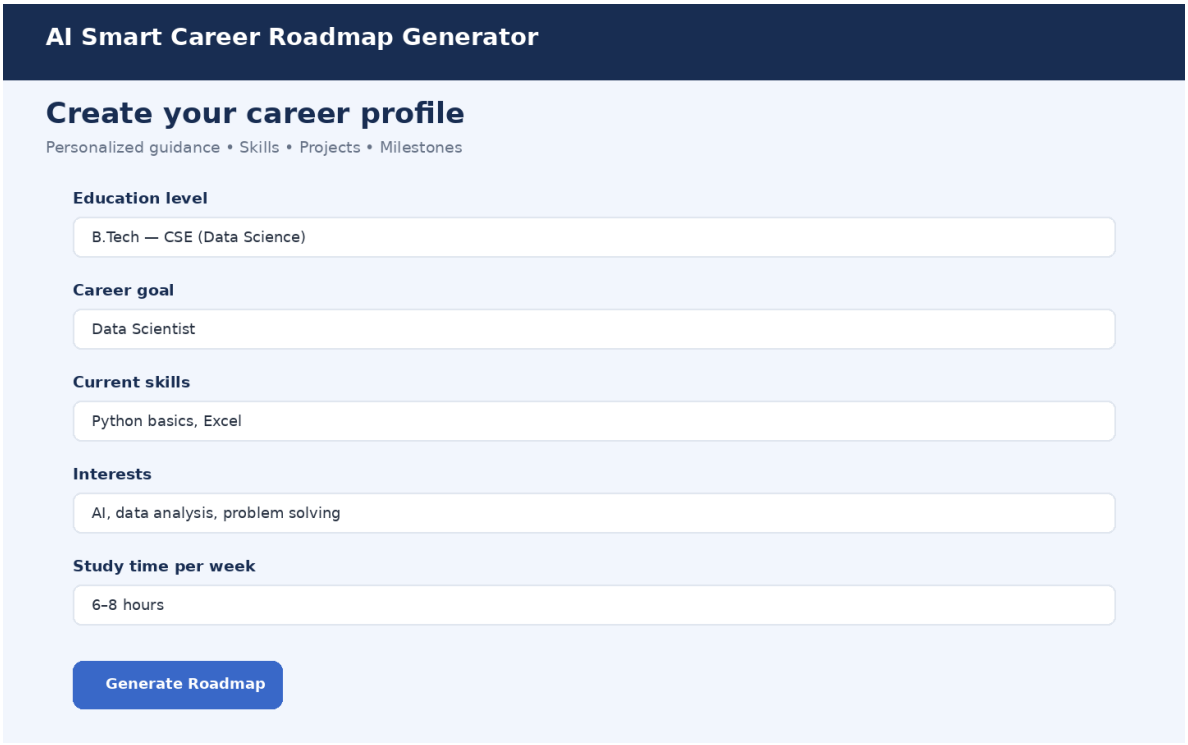


Figure 2: Career Profile Input


AI Smart Career Roadmap Generator
Create your career profile
Personalized guidance • Skills • Projects • Milestones
Education level
B.Tech — CSE (Data Science)
Career goal
Data Scientist
Current skills
Python basics, Excel
Interests
AI, data analysis, problem solving
Study time per week
6–8 hours
Generate Roadmap

8. User Interface / Application Screenshots (continued)

Figure 3: Personalized Career Roadmap

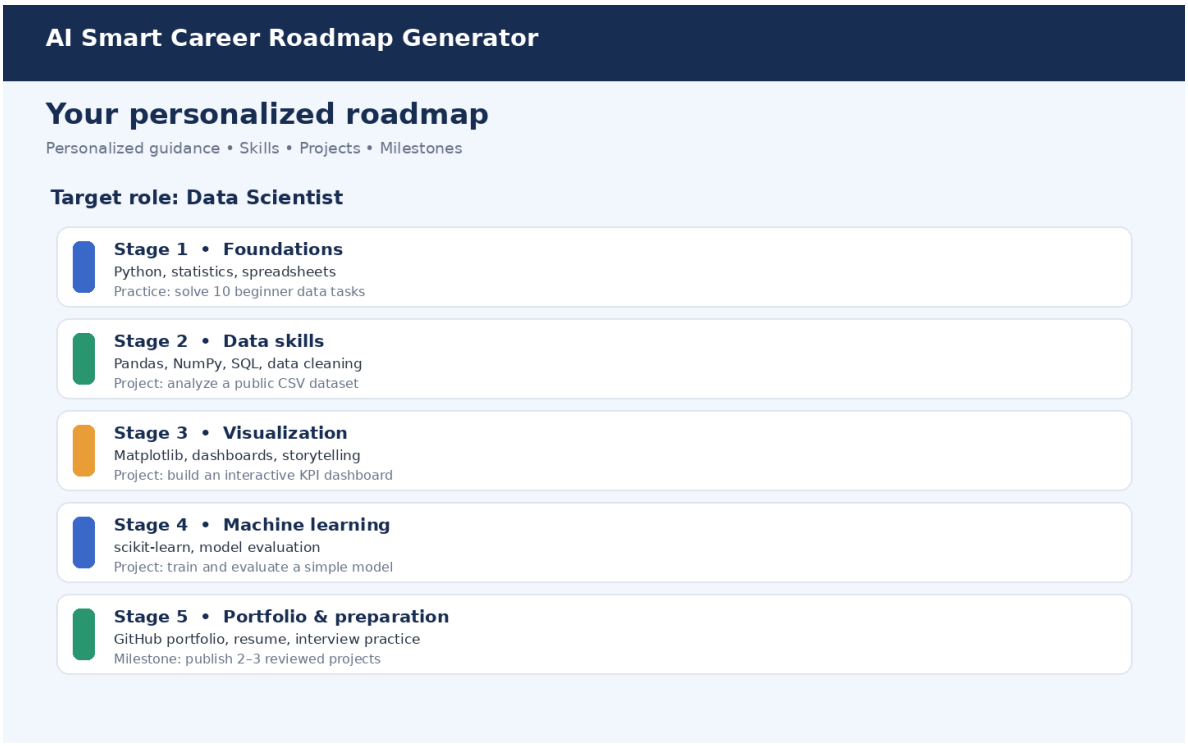
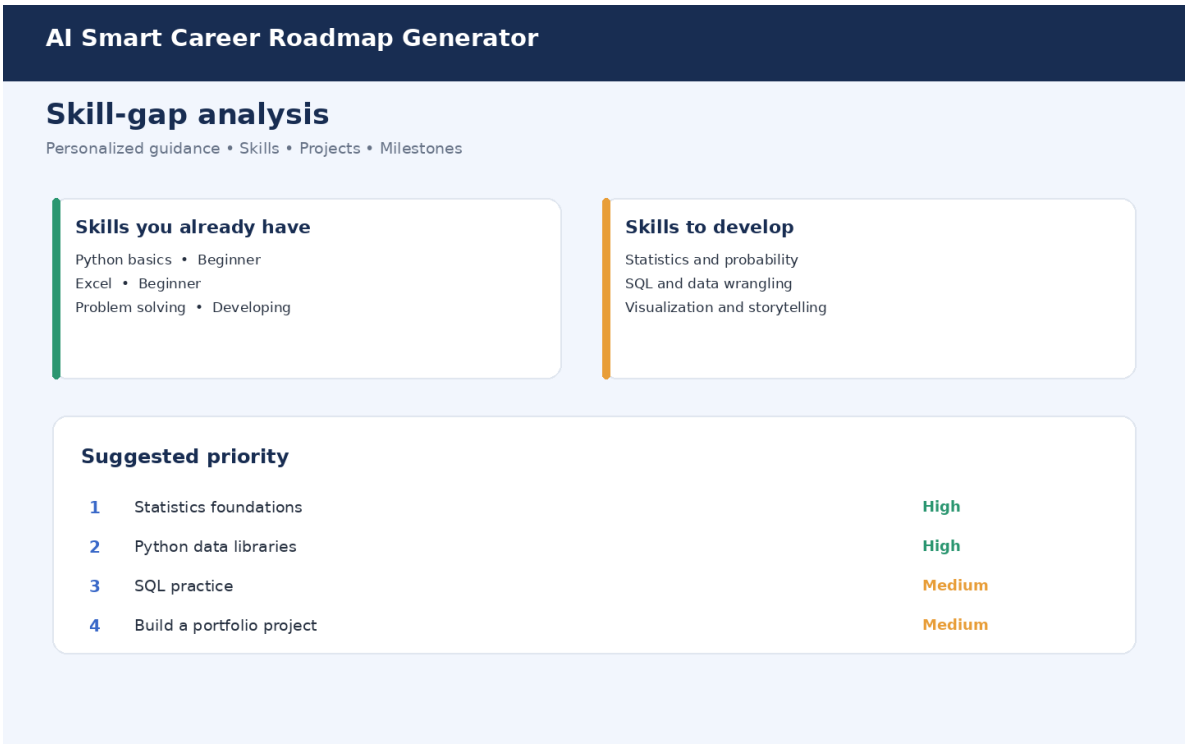


Figure 4: Skill-Gap Analysis



8. User Interface / Application Screenshots (continued)

Figure 5: Learning Progress Tracker

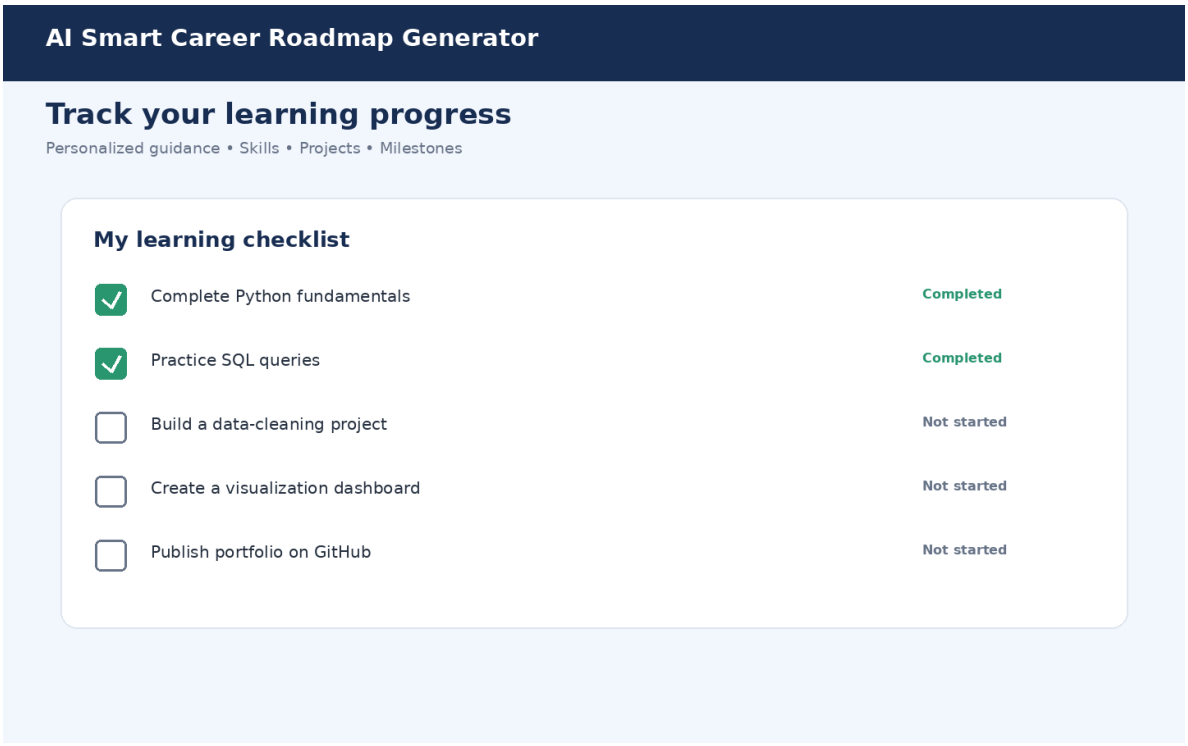
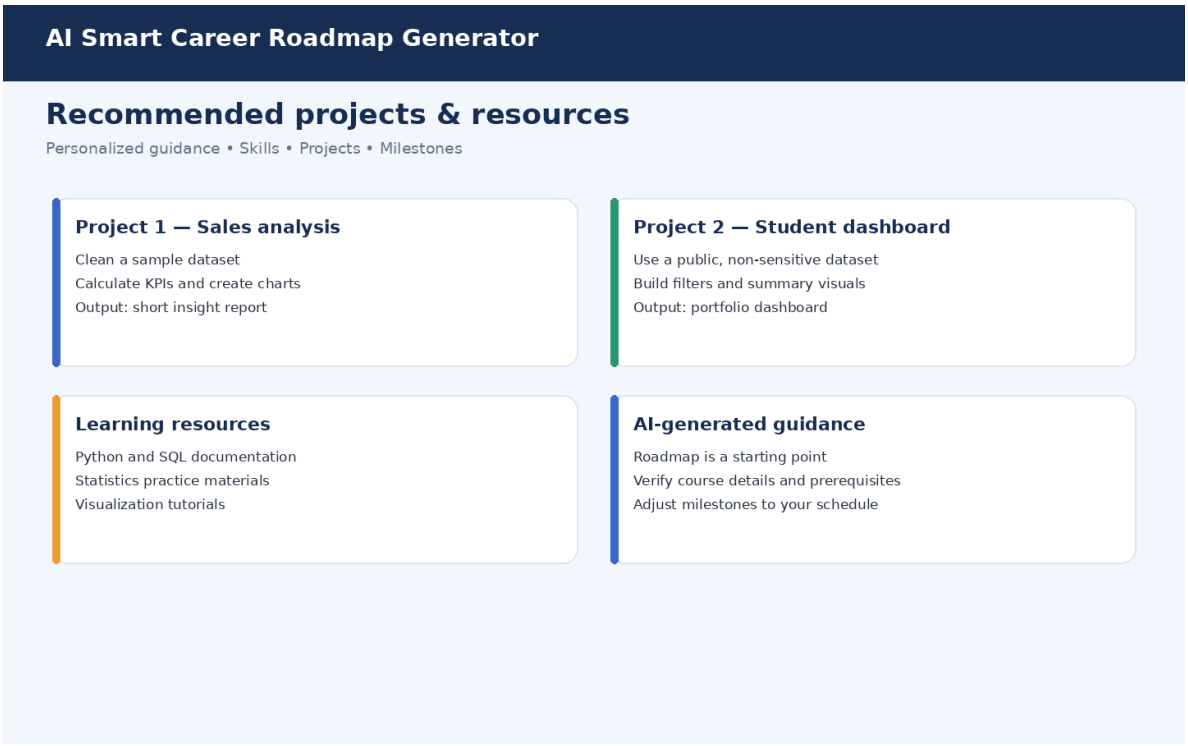


Figure 6: Projects & Learning Resources



9. Challenges and Limitations

Challenges

- Designing prompts that create clear, relevant, and realistic roadmaps.
- Keeping recommendations appropriate for different education and skill levels.
- Presenting a large amount of advice in a simple, readable dashboard.
- Keeping suggested learning resources and job requirements current.

Limitations

- AI-generated guidance may be incomplete or incorrect and should be verified.
- Recommendations depend on API availability, rate limits, and usage cost.
- The initial version may support English only and a limited set of career roles.
- A working implementation may need additional privacy, authentication, and testing controls.

10. Conclusion

The AI Smart Career Roadmap Generator is designed to make career planning more structured by turning a user's profile and career goal into staged learning guidance. The proposed features include career suggestions, skill-gap analysis, project recommendations, and a progress checklist.

The project offers an opportunity to learn prompt engineering, web development, API integration, response validation, and user-centered interface design. Actual performance claims should be added only after the application is implemented and evaluated.

11. Future Scope

- Add multilingual support for Indian languages.
- Connect to verified course and certification resources.
- Use current job descriptions to update skill requirements.
- Add roadmap export to PDF and resume-building features.
- Provide reminders, editable milestones, and improved progress analytics.
- Add secure login, privacy settings, and structured user feedback.
- Evaluate roadmap quality with a larger set of test profiles.

12. References

- Python documentation — https://docs.python.org/3/
- Flask documentation — https://flask.palletsprojects.com/
- FastAPI documentation — https://fastapi.tiangolo.com/
- MDN Web Docs — https://developer.mozilla.org/
- Official documentation for the Generative AI API actually used in the implementation.
- Relevant career information and course documentation consulted during development.

Note: The six application images in this report are illustrative UI mockups created for the project proposal, not screenshots captured from a running application. Replace them with real screenshots after implementation.